



Anil Neerukonda Institute of Technology & Sciences (Autonomous)

(Permanent Affiliation by Andhra University & Approved by AICTE
Accredited by NBA (ECE, EEE, CSE, IT, Mech. Civil & Chemical) & NAAC)

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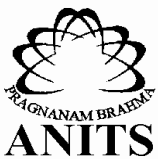
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Compiler Design Unit Wise Important Questions

UNIT-I

1	Define compiler? Explain Phases of compilation with examples?	[6M]	CO1	L1
2	Identify the Lexemes, Tokens and also write the number of tokens in a given program. main() { int *a,b; b=10; a=&b; printf("%d%d",b,*a); b=**pointer*/b; }	[6M]	CO1	L3
3	Explain the key elements in the lex program?	[6M]	CO1	L2
4	Explain about input buffering and its types.	[6M]	CO1	L2
5	Demonstrate the phases of compiler for the following assignment statement a: = (b +c)+12.	[12M]	CO1	L3
6	Compare PASS and PHASE with examples?	[6M]	CO1	L3
7	Explain Bootstrap technique with example?	[6M]	CO1	L2
8	Write a short notes on Tokens, Patterns and Lexemes. Give examples.	[6M]	CO1	L1&L3
9	Write a LEX program for count the vowels and consonants of the I/P string " MeN and WomeN".	[6M]	CO1	L6
10	What kind of errors can be detected during lexical analysis. Explain.	(4M)	CO1	L 2
11	Distinguish between Compiler and Interpreter	[6M]	CO1	L 2
12	Describe the need and Functionality of linker, assembler and loader.	[6M]	CO1	L2
13	Write a lex program to recognize keywords, identifiers and numbers.	[6M]	CO1	L3
14	Identify lexemes and tokens in the given code and count the number of tokens. 1int main() { double a, b; printf("Enter a: "); scanf("%lf", &a); printf("Enter b: ");	(6M)	CO1	L2

<pre>scanf("%lf", &b); a = a - b; b = a + b; a = b - a; printf("After swapping, a = %.2lf\n", a); printf("After swapping, b = %.2lf", b); return 0; }</pre>			
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UNIT-II

I	Check whether the following grammar is LL(1) or not $S \rightarrow iEtSS' a$ $S' \rightarrow eS \epsilon$ $E \rightarrow b$	[8M]	CO2	L3
2	Compare Top Down Parsing and Bottom up parsing with examples?	[4M]	CO2	L4
3	Explain Left Recursion and Left Factoring with an examples?	[6M]	CO2	L2
4	Explain Recursive Descent parser with an examples?	[6M]	CO2	L2
5	Eliminate the left recursion for the grammar . $A \rightarrow ABd Aa a$ $B \rightarrow Be b$	[6M]	CO2	L2
6	Check whether the following grammar is ambiguous or not. $S \rightarrow aS Sa \epsilon$	[6M]	CO2	L2
7	Show that the Following Grammar is LL(1) $S \rightarrow AaAb BbBa$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$	[8M]	CO2	L2
8	Check whether the following grammar is LL(1) or not $S \rightarrow iEtSS' a$ $S' \rightarrow eS \epsilon$ $E \rightarrow b$	[6M]	CO2	L3
9	Explain the recursive decent parser for the following grammar $S \rightarrow cAd$ $A \rightarrow ab/a$ and for the input "cad" trace the parser.	[6M]	CO2	L2
10	Eliminate left recursion in the following grammar $A \rightarrow ABcd Aa a$ $B \rightarrow Be b$	[6M]	CO2	L4
11	Explain FIRST() and FOLLOW() functions and compute FIRST() and FOLLOW() for the grammar $E \rightarrow E + T F$ $T \rightarrow T * F F$ $F \rightarrow (E) id$ Where the terminal set $T = \{+, -, (,), id\}$ and E is the start symbol.	[6M]	2	L2
12	Construct predictive parsing table for the following grammar. $S \rightarrow iEtSS' a$ $S' \rightarrow es \epsilon E \rightarrow b$ Determine whether the grammar is LL(1) or not.	(8M)	CO2	L3
13	Eliminate ambiguities in the following grammar.	[8M]	CO2	L3

	$S \rightarrow iEtS \mid iEtSeS \mid a$ $E \rightarrow b \mid c \mid d$			
14	Compute FIRST and FOLLOW sets for each non terminal of the grammar given below: $S \rightarrow ABa \mid bCa$ $A \rightarrow cBCD \mid \varepsilon$ $B \rightarrow CdA \mid ad$ $C \rightarrow eC \mid \varepsilon$ $D \rightarrow bSfa$	[8M]	CO2	L3
15	Construct Right most derivation for the grammar $E \rightarrow E+T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$ for $w = id + id * id$.	[8M]	CO-2	L6
16	Construct predictive parsing table for the grammar $E \rightarrow E+T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$	[8M]	CO-2	L6
17	Construct Predictive parsing table for the following grammar $E \rightarrow TE'$ $E' \rightarrow +TE' \mid \varepsilon$ $T \rightarrow FT'$ $T' \rightarrow *FT' \mid \varepsilon$ $F \rightarrow (E) \mid \varepsilon$	[15M]	CO2	L3



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Compiler Design Unit Wise Important Questions

UNIT-IV

1	What is meant by three address code? Different types of three address code techniques with examples?	[6M]	CO4	L1
2	Explain synthesized and inherited attributes with an example?	[6M]	CO4	L2
3	Construct quadruple, Triple and indirect triple for the following expression. $-(a+b)*(c+d)-(a+b+c)$?	[6M]	CO4	L3
4	What is meant by type checking ?with Examples	[6M]	CO4	L1
5	Develop an optimized code for while($i < 100$) { $a = \sin(x)/\cos(x) + i$; $i++$; }	[4M]	CO4	L2
6	Construct the expression for the SDD annotated parse tree for the following expressions $3 * 5 + 4n$.	[8M]	CO4	L3
7	Describe principle sources of optimization.	[6M]	CO4	L2
8	Translate the expression $(a+b)/(c+d)*(a+b/c)-d$ into quadruples, triples and indirect triples.	[6M]	CO4	L2
9	Explain Syntax Directed Translation With Example?	[6M]	CO4	L2
10	Describe the principle source of optimization with example?	[6M]	CO4	L3
11	What is a DAG? Construct DAG for the following basic block $d = b * c$; $e = a * b$; $b = b + c$; $c = b * c$;	[8M]	CO4	L1&L3
12	Describe Loop Optimization with an example.	[4M]	CO4	L2
13	Write a syntax directed definition to generate syntax tree for arithmetic expressions. $E \rightarrow E1 + T \quad \{E.nptr = mknode('+', E1.nptr, T.nptr) \}$	(6M)	CO4	L2



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Compiler Design Unit Wise Important Questions

UNIT-V

1	Write an algorithm for partition of basic blocks with an example?	[6M]	CO5	L3
2	What is meant by symbol table? Explain with an examples?	[6M]	CO5	L1
3	What is meant by PEEPHOLE OPTIMIZATION Techniques with an examples?	[6M]	CO5	L1
4	What is meant by machine dependent code optimization with examples?	[6M]	CO5	L1
5	What is an activation record? Explain how it is related with run time storage organization?	[6M]	CO5	L1
6	Explain about heap management?	[6M]	CO5	L2
7	Describe Register Allocation and Assignment.	[6M]	CO5	L2
8	What are the issues of code generation?	(4M)	CO5	L2
9	Discuss about the following: a) Copy Propagation b) Dead-code Elimination and c)Code motion.	[6M]	CO5	L2
10	Discuss in detail about Data flow analysis.	[6M]	CO5	L2
11	Write a note on i)strength reduction ii)Finding loop invariant iii)Loop unrolling iv)Constant folding v)Copy propagation	(6M)	CO5	L2
12	Optimize the code given below, by eliminating common sub expressions, performing reduction in strength on induction variables and eliminating all induction variables. dp=0 i=0 L: t1=i*8 t2=A[t1] t3=i*8 t4=B[t3] t5=t2*t4 dp=dp+5 i=i+1 If i<n goto L	[7M]	CO5	L3